

1. Administrative & Study Identification

Full Study Title: A Modular Phase I/II Open-label Dose Escalation and Dose Expansion Study to Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, and Efficacy of AZD5492, a T Cell-engaging Antibody Targeting CD20 in Subjects With Relapsed or Refractory B-Cell Malignancies (TITANium) **Short Title/Acronym:** TITANium **Protocol Number:** D9960C00001 **NCT Number:** NCT06542250 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD5492 (a first-in-class subcutaneous CD8-guided tri-specific T-cell engager / TCE) **Phase of Development:** Phase I/II **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability of AZD5492, determining the frequency of dose-limiting toxicities (DLTs). **Secondary Objectives:** To evaluate preliminary efficacy (Complete Response, Overall Response Rate, Duration of Response, PFS, OS), pharmacokinetics (PK), pharmacodynamics, and immunogenicity (Anti-Drug Antibodies) of AZD5492.

2.2 Background & Rationale To address the limitations of bispecific CD20 x CD3 T-cell engagers (TCEs) in relapsed or refractory non-Hodgkin lymphomas. While TCEs are changing the treatment landscape, they are associated with severe immune-related toxicities, namely cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS). AZD5492 is designed as a CD8-guided tri-specific TCE that may offer a wider therapeutic index and safety advantage by specifically targeting CD8+ T cells to reduce cytokine release while maintaining potent antitumor activity.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose escalation and Dose expansion modules) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 174 **Number of Sites:** Global Multicenter **Estimated Study Start:** 18-Sep-2024 **Estimated Primary Completion:** 14-Feb-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. ≥ 18 years of age. 2. Histologically documented CD20+ mature B-cell neoplasm (Large B-cell lymphoma, Follicular lymphoma, Mantle cell lymphoma, Chronic lymphocytic leukemia, or Small lymphocytic lymphoma). 3. Relapsed, progressive and/or refractory disease following at least 2 prior lines of therapy. 4. ECOG performance status of ≤ 2 (< 2 in EU countries).

4.2 Exclusion Criteria 1. Active CNS involvement in lymphoma or CNS pathology (e.g., history of seizure disorder/epilepsy). 2. Diagnosis of post-transplant lymphoproliferative disease, Richter's transformation, Burkitt's lymphoma, or Burkitt-like lymphoma. 3. History of Grade ≥ 3 CRS or Grade ≥ 3 ICANS from prior therapies. 4. Prior allogeneic HSCT within 180 days, prior autologous HSCT within 90 days, or cell therapy within 90 days of start of therapy.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered at a fixed dose per dose-level, utilizing an accelerated titration and step-up dosing strategy if triggered by immune-related toxicities. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** Until disease progression, unacceptable toxicity, or completion of the defined treatment modules (up to 2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing immune-related side effects. **Prohibited:** Other concurrent anticancer therapies unless specified under combination modules.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Tumor Assessments
Baseline	Day 1	First Dose of AZD5492, PK/PD Sampling, Safety Monitoring
Interim	Cycle 1 - 5 Weeks	Intensive safety tracking for DLTs, CRS, and ICANS monitoring
End of Study	Up to 2 years	Follow-up for PFS, OS, and ADA evaluation at 90 days post last dose

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Frequency of dose-limiting toxicities (DLTs) from the first administration of AZD5492 until the end of cycle 1 (up to 5 weeks), and incidence of treatment-

related adverse events. **Measurement Method:** Clinical safety evaluation per protocol guidelines and CTCAE criteria.

7.2 Secondary & Exploratory Endpoints * Proportion of participants with Complete Response (CR) and Overall Response Rate (ORR) according to Lugano criteria (malignant lymphoma) or iwCLL 2018 criteria (CLL). * Duration of Response (DoR), Progression-Free Survival (PFS), Overall Survival (OS). * Pharmacokinetics (Cmax) and Immunogenicity (Anti-Drug Antibodies [ADA]).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Intensive monitoring for specific immune-related toxicities including Cytokine Release Syndrome (CRS) and Immune effector cell-associated neurotoxicity syndrome (ICANS). **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline. **Data Monitoring Committee (DMC):** Safety review committee utilizing a modified toxicity probability interval-2 (mTPI-2) design to guide escalation and dose expansion.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (for evaluating AEs/DLTs) and Efficacy Evaluable Set. **Sample Size Justification:** Target N=174, with dose escalation governed by accelerated titration and modified toxicity probability interval-2 (mTPI-2) design. **Handling of Missing Data:** Standard survival analysis right-censoring methods for time-to-event endpoints (PFS/OS) and imputation for PK/PD values where applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical monitoring for safety and protocol compliance. **Audit Trail:** eCRF data capture with specific audit trails for step-up dose administration and toxicity event tracking.

11. Study Identifiers & Governance

Primary ID: D9960C00001 **Registry Number:** NCT06542250 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** TITANium **Official Regulatory Title:** A Modular Phase I/II Open-label Dose Escalation and Dose Expansion Study to Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, and Efficacy of AZD5492, a T

Cell-engaging Antibody Targeting CD20 in Subjects With Relapsed or Refractory B-Cell Malignancies

(Note: The following sections have been adapted from the provided template to fit the specific oncology framework of NCT06542250).

12. Clinical Framework (Oncology Specific)

Primary Condition: Relapsed or Refractory B-Cell Malignancies
Diagnosis Threshold: Histologically documented CD20+ mature B-cell neoplasm **Medical Methodology:** Subcutaneous CD8-guided tri-specific T-cell engager (TCE) **Optimization Target:** Maximum Tolerated Dose (MTD) / Recommended Phase 2 Dose (RP2D)

13. Study Procedures & Methodology

Management Pathway: Monotherapy dose escalation with step-up dosing strategies **Education Protocol (HCP):** Identification and management protocols for CRS and ICANS **Education Protocol (Patient):** Specific side-effect reporting and symptom monitoring for immunotherapy **Quality Audit Mechanism:** Continuous safety reviews (mTPI-2) for DLT incidence

14. Observation & Follow-Up Schedule

Total Duration: Up to 2 years after last dose **Onsite Visit Cadence:** Module-specific treatment cycles starting at Baseline and Interim (Cycle 1) **Remote Monitoring Frequency:** Post-treatment survival follow-up **Checklist Review Interval:** Rigorous DLT monitoring during the first 5 weeks of dosing

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Insert Name]			____		[DD-MMM-YYYY]							
	Clinical Operations Lead				[Insert Name]			____		[DD-MMM-YYYY]							
	Lead Statistician				[Insert Name]			____		[DD-MMM-YYYY]							